Closed-form Option Formulas for Kou-like Models*

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Abstract

The aim of this study is to expand the Kou model to emulate the Samuelson effect, a characteristic attribute observed in commodity options. This study explores both the NOA additive jump diffusive model for future contracts and an exponential jump diffusion dynamics. We derive closed-formulas for call and put options with different types of jump distributions all related to the exponential law. As a side product, we can derive the densities of the convolution between several Erlang-like laws and the Gaussian distribution in addition to simplifications of the original Kou model formulas, rendering them more accessible involving simple functions only. Finally, we present extensive numerical applications within energy markets and conduct comparative analyses against other well-established models.

1 Introduction and Motivation

The energy market serves as the bedrock of modern civilization, encompassing various sectors pivotal to daily life, industry, and global economies. As societies continue to evolve, so do the demands and complexities within the energy landscape. The energy market is a multifaceted ecosystem comprising diverse sources, including fossil fuels, renewable energy, nuclear power, and emerging technologies. However, it faces a multitude of challenges in meeting global energy demands while transitioning towards sustainability.

One important aspect is that electricity is not storable in the traditional sense akin to classical commodities. This circumstance holds significant implications.

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Notably, in the realm of electricity futures contracts, the underlying asset is not delivered at a singular point in time but across specified delivery periods of varying durations. These periods commonly span monthly, quarterly, and yearly intervals, defining the distinct nature of electricity market contracts.

Energy markets are inherently subject to volatility, influenced by geopolitical tensions, global supply chains, and shifts in demand patterns. This volatility often leads to price fluctuations and supply disruptions, posing challenges for the mathematical modelling.

Recent literature has placed significant emphasis on modeling techniques rooted in non-Gaussian processes (see for instance Benth et al. [2], and Benth et al. [3], Latini et al. [21] and Gardini et al. [12] among others). This approach aims to encompass specific characteristics observed in real market data, such as jumps, heavy tails, and asymmetry. Particularly noteworthy is the period surrounding and following the Russian-Ukrainian conflict, during which commodity prices exhibited heightened volatility and irregularities that defy conventional Gaussian modeling. This erratic behavior underscores the limitations of Gaussian models in capturing the complexities and irregularities present in real-world market dynamics.

Moreover, commodity option prices exhibit distinctive traits including skew, seasonality, and what is known as the Samuelson effect (see for instance Jaeck and Lautier [16], Kiesel et al. [17], Benth and Schmeck [4] and Fanelli and Schmeck [10] among others). This effect denotes the phenomenon where volatility increases as maturities draw closer, highlighting a noteworthy pattern observed within commodity markets.

In particular Piccirilli et al. [26] have recently introduced a set of models termed non-overlapping-arbitrage models (NOA), designed specifically to account for the Samuelson effect. These models aim to replicate the diverse implied volatility profiles exhibited by options, thereby extending the understanding and capturing the essence of this phenomenon.

The modeling of the Samuelson effect using Lévy processes is closely linked to the study of Lévy-driven Ornstein–Uhlenbeck (OU) processes. For example, the original model proposed by Piccirilli et al. [26] is based on Normal Inverse Gaussian (NIG) processes, or more precisely on OU-NIG processes, following the notation introduced by Barndorff–Nielsen and Shephard [1]. More recent developments have extended this framework to encompass broader classes of OU processes. In particular, Sabino [30, 32] investigates OU processes driven by normal tempered stable (NTS) Lévy noise, while Sabino [29, 31] and Goutte et al. [13] focus on OU models based on Variance Gamma (VG) and CGMY processes. These latter works explicitly demonstrate the link between models for spot price dynamics in energy markets and the forward price dynamics that captures the Samuelson effect.

In the present study, we focus on extending the well-known Kou model [20] in several directions, thus contributing to the further development of this class of models. Specifically, our aim is to introduce a novel alternative to those already available in the literature.

Importantly, in contrast to much of the existing literature on forward price modeling, and particularly stepping beyond the Gaussian framework, our approach provides a comprehensive analytical toolkit which includes the density function of the price process, closed-form expressions for vanilla option prices which, unlike in the NTS, VG and CGMY cases, are available in closed form, and the characteristic function of the process. Furthermore, the simulation methods presented in Sabino and Cufaro Petroni [33, 34] are incorporated as part of this framework.

Hence, in contrast to the existing models one has a comprehensive toolkit for parameter estimation, using historical and option data as well as several methods for option pricing, namely Monte Carlo (MC) simulations and FFT-based numerical techniques.

The contribution of our study is manifold. As an initial result, our study establishes the convolution density arising from the convolution of the Gaussian distribution with diverse Erlang-like distributions, encompassing the bilateral case. To our knowledge, the statistical literature solely presents a direct formulation for the convolution density resultant from combining Gaussian and exponential distributions, known as the exponentially-modified Gaussian distribution see (Grushka [15] and Delley [9]). This outcome holds significance for parameter estimation, although it requires further exploration. Specifically, opting for a simple maximum log-likelihood estimation method may not be advisable. However, alternative techniques such as the Expectation-Maximization (EM) method or stochastic gradient descent might be more suitable. These methods are computationally less expensive compared to Monte Carlo (MC)-based methods commonly employed for this purpose (see Laudagé et al. [22]). We emphasize that none of the density formulas necessitate numerical integration or approximation techniques.

This leads to the second contribution, which involves simplifying the option formulas in the original Kou model that initially comprised complex confluent hypergeometric functions.

Additionally, the third contribution entails deriving closed-formulas for vanilla options, resembling the structure of the Merton [25] and Kou models, expressed in terms of infinite sums. We utilize a convenient truncation method based on the findings of Sabino and Cufaro Petroni [34].

We validate our findings through extensive numerical simulations, comparing the results with Monte Carlo (MC) methods. Finally, we validate the extended Kou model through empirical calibration using observed implied volatilities derived from settlement prices. The results are compared with those obtained from a OU-modified version of the VG model, adapted to account for the Samuelson effect, as proposed in Sabino [29] and Goutte et al. [13].

The paper is structured as follows. Section 2 establishes the notation the notation while Section 3 introduces the extended Kou multi-factor model in the context of energy markets. In Section 4 we derive the density function of several Erlang-like modified Gaussian distribution. Then Section 5 presents the closed-formulas for vanilla options corresponding to various Kou-like market models. Section 6 conducts a numerical comparison between option values obtained using closed-formulas and MC simulations. Furthermore, Section 7 details the application to real option data on German power futures, and lastly, Section 8 concludes the paper.

2 Notation

In this section we introduce the notation for known distributions and the shortcuts used throughout the paper.

We write $\mathcal{N}(\mu, \sigma^2)$, $\sigma > 0$, to denote the Gaussian distribution with mean μ and variance σ^2, and density function (*pdf*) $g(\cdot\,|\mu, \sigma)$, cumulative distribution function (*cdf*) $\Phi(\cdot\,|\mu, \sigma)$, and survival function $\bar{\Phi}(\cdot\,|\mu, \sigma)$ respectively. For the standard Gaussian law, we will omit μ and σ. We denote by $\mathcal{P}(\lambda)$ the Poisson distribution with parameter $\lambda > 0$ and by $\overline{\mathfrak{B}}(\alpha, p)$ the Pólya distribution with parameter $\alpha > 0$ and probability of success p, namely

$$\boldsymbol{P}\left\{X = k\right\} = \binom{\alpha + k - 1}{k}(1-p)^{\alpha}p^{k}, \quad k = 0, 1, \dots \tag{1}$$

where

$$\binom{\alpha + k - 1}{k} = \frac{\alpha(\alpha-1)\cdot(\alpha-k+1)}{k!}, \binom{\alpha}{0} = 1.$$

We refer to the Erlang distribution with shape parameter n and rate λ with $\mathcal{E}_n(\lambda)$. We recall that the Erlang law is the special case of the gamma law $\mathcal{G}(\alpha, \lambda)$ ($\alpha > 0, \lambda > 0$) with $\alpha = n \in \mathbb{N}^*$ having the following *pdf* and characteristic function (*chf*) (see Sabino and Cufaro Petroni [34]):

$$f(x|n, \lambda) = \frac{\lambda}{(n-1)!}(\lambda x)^{n-1}e^{-\lambda x} \qquad x > 0, \tag{2}$$

$$\varphi(u, |n, \lambda) = \left(\frac{\lambda}{\lambda - iu}\right)^{n}. \tag{3}$$

We denote by $b\,\mathcal{E}_n(\lambda)$ the bilateral Erlang law, for $n = 1$ which can be seen as the difference $X - Y$ of two independently and identically distributed (*iid*) random variables (*rv*'s) $X, Y \sim \mathcal{E}_n(\lambda)$. We recall that the bilateral Erlang law has the following *pdf* and *chf* (see Sabino and Cufaro Petroni [34]):

$$\bar{f}(x|n, \lambda) = \frac{\lambda}{2^n(n-1)!}\left(\lambda|x|\right)^{n-1}e^{-\lambda|x|}\sum_{k=0}^{n-1}\frac{(n-1+k)!}{k!(n-1-k)!(2\lambda)^k|x|^k}, \tag{4}$$

$$\bar{\varphi}(u, |n, \lambda) = \left(\frac{\lambda^2}{\lambda^2 + u^2}\right)^{n}. \tag{5}$$

To ease the notation we define :

$$\omega_{k,n} = \frac{(n-1+k)!}{k!(2\lambda)^k} \tag{6}$$

and rewrite Equation (4) as follows :

$$\bar{f}(x|n,\lambda) = \frac{\lambda^n}{2^n(n-1)!} e^{-\lambda|x|} \sum_{k=0}^{n-1} \frac{\omega_{k,n}}{(n-1-k)!} |x|^{n-k-1}. \tag{7}$$

In addition, we recall that the *pdf* and the *cdf* of the difference between two independent Erlang *rv*'s, $\mathcal{E}_n(\lambda_u)$ and $\mathcal{E}_m(\lambda_d)$, are (see Simon [37] page 28)

$$f(x|n,m,\lambda_u,\lambda_d) = \begin{cases} \frac{\lambda_d e^{\lambda_d x}}{(n-1)!} \left(\frac{\lambda_u}{\lambda_u+\lambda_d}\right)^n \sum_{i=0}^{m-1} \frac{(n+m-i-2)!}{i!(m-i-1)!} \times \\ \left(\frac{\lambda_d}{\lambda_u+\lambda_d}\right)^{m-i-1} (-\lambda_d x)^i \quad x < 0, \\ \frac{\lambda_u e^{-\lambda_u x}}{(m-1)!} \left(\frac{\lambda_d}{\lambda_u+\lambda_d}\right)^m \sum_{j=0}^{n-1} \frac{(n+m-j-2)!}{j!(n-j-1)!} \times \\ \left(\frac{\lambda_u}{\lambda_u+\lambda_d}\right)^{n-j-1} (\lambda_u x)^j \quad x \geq 0 \end{cases}, \tag{8}$$

$$\varphi(u|n,m,\lambda_u,\lambda_d) = \left(\frac{\lambda_u}{\lambda_u - iu}\right)^n \left(\frac{\lambda_d}{\lambda_d + iu}\right)^m. \tag{9}$$

Equation (7) is a special case of Equation (8), the former one can be obtained with $\lambda = \lambda_u = \lambda_d$, $n = m$ and setting $j = i = k$. It is then convenient to rewrite Equation (8) setting $\ell = m - i - 1$ and $k = n - j - 1$ with

$$\alpha_{\ell,n} = \frac{(n+\ell-1)!}{\ell!} (\lambda_u + \lambda_d)^{-\ell} \tag{10}$$

for which we obtain:

$$f(x|n,m,\lambda_u,\lambda_d) = \begin{cases} \frac{\lambda_d^m e^{\lambda_d x}}{(n-1)!} \left(\frac{\lambda_u}{\lambda_u+\lambda_d}\right)^n \sum_{\ell=0}^{m-1} \frac{\alpha_{\ell,n}}{(m-\ell-1)!} (-x)^{m-\ell-1} \quad x < 0, \\ \frac{\lambda_u^n e^{-\lambda_u x}}{(m-1)!} \left(\frac{\lambda_d}{\lambda_u+\lambda_d}\right)^m \sum_{k=0}^{n-1} \frac{\alpha_{k,m}}{(n-k-1)!} x^{n-k-1} \quad x \geq 0 \end{cases}, \tag{11}$$

3 Kou-like models and option formulas

The original Kou model (see Kou [20]) provides closed-form formulas for plain vanilla options under the assumption that the dynamic of the price $S(t)$ a risky asset is driven by a jump-diffusion process

$$\log S(t) = \log S(0) + \left(\mu - \frac{1}{2}\sigma^2\right) t + \sigma W(t) + \sum_{n=1}^{N(t)} J_n \tag{12}$$

where $J_n, n > 0$ are *iid* copies following a double exponential law with *pdf*

$$f_J(x) = p\lambda_u e^{-\lambda_u x} \mathbb{1}_{x\geq 0} + (1-p)\lambda_d e^{\lambda_d x} \mathbb{1}_{x<0} \tag{13}$$

and $N(\cdot)$ is a Poisson process with intensity λ_P. With little effort the model can be adapted to the case where the jumps follow a different exponential-type law indeed, $f_J(\cdot)$ can also be seen as mixture of a positive and a negative exponential distributions. In addition, we assume that the diffusive and jump components are independent.

In this study we extend the Kou model allowing Equation (12) to have time-dependent parameters to accommodate the Samuelson effect and the volatility smile, namely we take an exponential model similar to Kiesel et al. [17]. We start considering the dynamics of a forward contract with maturity T and delivery period $[T_1, T_2], T \leq T_1$ as follows:

$$\log F(t, T_1, T_2) = \log F(0, T_1, T_2) + h(t, T_1, T_2) + \int_0^t \Gamma_1(u, T_1, T_2) dL_1(u) + \Gamma_2(T_1, T_2) W_2(t)$$

where $L_1(t) = W_1(t) + \sum_{n=1}^{N(t)} J_n$. $W_1(\cdot)$ and $W_2(\cdot)$ are two independent Brownian motions (BM's). Of course, this last assumption can be easily relaxed. Moreover, in order to capture the Samuelson effect it is customary to take

$$\Gamma_1(t, T_1, T_2) = \frac{\gamma_1 \left(e^{-k(T_1 - t)} - e^{-k(T_2 - t)}\right)}{k(T_2 - T_1)}, \tag{14}$$

$$\Gamma_2(T_1, T_2) = \frac{1}{T_2 - T_1} \int_{T_1}^{T_2} \gamma_2(u) du, \tag{15}$$

where γ_1 and γ are two strictly positive constants The time-dependent function $h(\cdot, T_1, T_2)$ is such that the process $F(\cdot, T_1, T_2)$ is a martingale. Clearly here, and hereafter, we assume that we live in a risk-neutral world.

We consider however, a slightly different setting and split the impact Γ_1 into a diffusive and a pure jump part

$$\Sigma_p(u, T_1, T_2) = \frac{\gamma_p \left(e^{-k_p(T_1 - u)} - e^{-k_p(T_2 - u)}\right)}{k_p(T_2 - T_1)}, p \in \{D, J\}. \tag{16}$$

We could take $\gamma_D = \gamma_J$ and $k_D = k_J$ reducing the number of parameters which could be done nevertheless in a second step. Accordingly, it is easy to rewrite the equation of the forward dynamics as

$$\begin{aligned} \log F(t, T_1, T_2) = &\log F(0, T_1, T_2) + h(t, T_1, T_2) + \Gamma_2(T_1, T_2) W_2(t) \\ &+ \Sigma_D(t, T_1, T_2) Z_D(t) + \Sigma_J(t, T_1, T_2) Z_J(t) \end{aligned} \tag{17}$$

$$Z_D(t) = \int_0^t e^{-k_D(t-u)} dW_1(u), \qquad Z_J(t) = \sum_{n=1}^{N(t)} J_n e^{-k_J(t - \tau_n)}.$$

It is worth noting that modifying the original jump-diffusion model proposed by Merton [25] to incorporate the Samuelson effect is not a straightforward task. This adjustment becomes complex as the jumps are assumed to be normally distributed, since the *chf* and the *pdf* of $Z_J(\cdot)$ to our knowledge are not known in simple form.

Alternatively, we also consider an additive framework in the spirit of the NOA models introduced in Piccirilli et al. [26]. To this end, we recall that the NOA model for energy forward, in its simple form with just two risk factors, is [1]

$$F(t,T_1,T_2) = F(0,T_1,T_2)(1+h(t,T_1,T_2)+\int_0^t \Gamma_1(u,T_1,T_2)dL_1(u)+\Gamma_2(T_1,T_2)L_2(t)) \tag{18}$$

where $L_i(\cdot), i=1,2$ are two Lévy processes potentially dependent. Following the same ideas of the exponential model can rewrite equation (18) as follows

$$\begin{aligned}
F(t,T_1,T_2) =& F(0,T_1,T_2)(1+h(t,T_1,T_2)+\Gamma_2(T_1,T_2)W_2(t)+ \\
&+\Sigma_D(t,T_1,T_2)\int_0^t e^{-k_D(t-u)}dW_1(u)+\Sigma_J(t,T_1,T_2)\sum_{n=1}^{N(t)} J_n e^{-k_J(t-\tau_n)}) \\
=& F(0,T_1,T_2)(1+h(t,T_1,T_2)+\Gamma_2(T_1,T_2)W_2(t) \\
&+\Sigma_D(t,T_1,T_2)Z_D(t)+\Sigma_J(t,T_1,T_2)Z_J(t)).
\end{aligned} \tag{19}$$

Apparently, $Z_p(\cdot), p \in \{J,D\}$, is neither a Lévy nor an additive process however, at any fixed time t, the law of the *rv* $Z_p(t)$ is *id* therefore, one can use the machinery of the Lévy-Khinchine representation theorem. In addition, such a process always pops up for Ornstein-Uhlenbeck processes and it has been extensively studied in a series of papers (see Sabino and Cufaro Petroni [34, 35], Sabino [29, 30, 32], Grabchack and Sabino [14], Qu et al [27, 28]) encompassing several distributions.

Finally, we will take different exponential-like distributions for the jumps namely, simple exponential, Laplace, and bilateral exponential laws, but we will also consider a jump component with two independent terms describing positive and negative exponential jumps $Z_J(t) = Z_u(t) - Z_d(t)$.

In a risk-neutral world, the price of a European-style derivative contract depending on the forward with maturity T, delivery period $[T_1, T_2]$ and payoff $\phi(\cdot)$ is given by $\boldsymbol{E}[\phi(F(T,T_1,T_2))]$ therefore, for to market model in Equation (17) we have that at time T

$$\begin{aligned}
\log F(T,T_1,T_2) &\stackrel{d}{=} \log F(0,T_1,T_2)+h(T,T_1,T_2)+X+\Sigma_J(T,T_1,T_2)\,Z \\
&= \log F(0,T_1,T_2)+h(T,T_1,T_2)+E,
\end{aligned} \tag{20}$$

where $X \sim \mathcal{N}(0,\sigma^2(T,T_1,T_2))$, $\sigma^2(T,T_1,T_2) = \Sigma_D^2(T,T_1,T_2)\left(\frac{1-e^{-2k_D T}}{2k_D}\right)+\Gamma_2(T_1,T_2)^2$ and $Z \stackrel{d}{=} \sum_{n=1}^{N(T)} J_n e^{-k(T-\tau_n)}$. As mentioned above, we will also cover the case where Z has the same law of the difference of two independent jump processes, we omit it for the moment for brevity.

[1] The model is slightly different than the original one which is $F(t,T_1,T_2) = F(0,T_1,T_2) + h(t,T_1,T_2) + \int_0^t \Gamma_1(u,T_1,T_2)dL_1(u) + \Gamma_2(T_1,T_2)L_2(t)$, we prefer to keep the random terms and $h(t,T_1,T_2)$ dimensionless.

For call and put options the payoff functions are $(F(T,T_1,T_2)-K)^+$ and $(K-F(T,T_1,T_2))^+$, respectively, where K denotes the strike price, for simplicity we take the risk-free rate r equal to zero. Therefore, the call option price at time $t=0$ is

$$\begin{aligned}
C(0) &= \boldsymbol{E}\left[F(T,T_1,T_2)\mathbb{1}_{F(T,T_1,T_2)\geq K}\right] - K\boldsymbol{P}\left\{F(T,T_1,T_2)\geq K\right\} \\
&= e^{h(T,T_1,T_2)}F(0,T_1,T_2)\Psi_1\left(\log\frac{K}{F(0,T_1,T_2)} - h(T,T_1,T_2)\right) \\
&- K\Psi_2\left(\log\frac{K}{F(0,T_1,T_2)} - h(T,T_1,T_2)\right)
\end{aligned} \tag{21}$$

where $f_E(\cdot)$ denotes the *pdf* of E and

$$\Psi_1(c) = \int_c^{+\infty} e^x f_E(x)dx \tag{22}$$

$$\Psi_2(c) = \int_c^{+\infty} f_E(x)dx, \tag{23}$$

$\Psi_2(\cdot)$ is apparently the survival probability function of E. The put option price can be retrieved by put-call parity. $h(T,T_1,T_2)$ depends on the cumulants of E and X

$$h(T,T_1,T_2) = -\frac{\sigma^2(T,T_1,T_2))}{2} - m_E(1), \tag{24}$$

where $m_E(\cdot)$ is the cumulant generating function (*cgf*) of E. The last result is instrumental for the pricing and calibration of call and put options using FFT-based methods (see for instance Carr and Madan Carr [6]).

On the other hand, instead of relying on FFT methods, one could price such options using closed-formulas provided the *pdf* of E be known. To this end, Sabino and Cufaro Petroni [34] observed that the law of Z is an Erlang-like mixture distribution where the mixing law is a Pólya distribution. Therefore, the *pdf* of E (and the integrals Ψ_1 and Ψ_2) can be written as a convex sum $f_E(x) = \sum_{n=0}^{\infty} p_n f_{E_n}(x)$, where $f_{E_n}(\cdot)$ is the *pdf* of the convolution law between a Gaussian and an Erlang-like law and p_n are weights determined by the Pólya mixture.

Finally, these observations are also applicable to the NOA dynamics, thus

$$\begin{aligned}
C(0) =&\boldsymbol{E}\left[(F(T,T_1,T_2)-K)^+\right] = \boldsymbol{E}\left[(F(0,T_1,T_2)(1+h(T,T_1,T_2)+E)-K)^+\right] \\
=&F(0,T_1,T_2)\,\Xi\left(\frac{K}{F(0,T_1,T_2)} - 1 - h(T,T_1,T_2)\right) \\
&-(K-F(0,T_1,T_2)(1+h(T,T_1,T_2)))\,\Psi_2\left(\frac{K}{F(0,T_1,T_2)} - 1 - h(T,T_1,T_2)\right)
\end{aligned} \tag{25}$$

where

$$\Xi(c) = \int_c^{+\infty} x\, f_E(x)dx \tag{26}$$

and now $h(T, T_1, T_2) = -\boldsymbol{E}[E]$.

In the following section we will derive the *pdf*'s of the convolution law between a Gaussian and some Erlang-like laws namely, Erlang, bilateral Erlang and difference of two independent Erlang distributions. These results will be instrumental to derive closed formulas for call and put options that will presented in Section 5, but also to implement parameter estimation routines.

4 Convolution of a Gaussian and an Erlang-type law

In this section we derive the *pdf*'s of the sum of a Gaussian *rv* $X \sim \mathcal{N}(\mu, \sigma)$ plus an Erlang-type *rv*. Moreover, we compute some convenient integrals for the derivation of the option price formulas.

4.1 Convolution of a Gaussian and an Erlang law

Consider the *rv* $X + Y$ where $Y \sim \mathcal{E}_n(\lambda)$. The *pdf* is obtained via convolution $s(x|\mu, \sigma, n, \lambda) = \int_{\mathbb{R}^+} g(x - y|\mu, \sigma) f_{n,\lambda}(y) dy$.

$$s(x|\mu, \sigma, n, \lambda) = \frac{\lambda^n}{\sqrt{2\pi}\sigma(n-1)!} \int_{\mathbb{R}^+} y^{n-1} \exp\left(-\frac{(x-y-\mu)^2 + 2\sigma^2\lambda y}{2\sigma^2}\right) dy$$

Now we focus on the argument of the exponential

$$A_u(x, y|\lambda, \sigma, \mu) = (x - y - \mu)^2 + 2\sigma^2\lambda y,$$

and on the quantity

$$(x - y - \mu - \sigma^2\lambda)^2 = (x - y - \mu)^2 + \sigma^4\lambda^2 - 2\sigma^2\lambda(z - y - \mu).$$

With some simple algebra we have

$$A_u(x, y|\lambda, \sigma, \mu) = (x - y - \mu - \sigma^2\lambda)^2 - \sigma^4\lambda^2 + 2\sigma^2\lambda(x - \mu),$$

then

$$s(x|\mu, \sigma, n, \lambda) = \frac{\lambda^n}{\sqrt{2\pi}\sigma(n-1)!} e^{E_u(x|\lambda,\sigma,\mu)} \int_{\mathbb{R}^+} y^{n-1} \exp\left(-\frac{(x-y-\mu-\sigma^2\lambda)^2}{2\sigma^2}\right) dy,$$

where $E_u(x|\lambda, \sigma, \mu) = -\lambda x + (\sigma^2\lambda^2 + 2\mu\lambda)/2$. With the change of variable $\tau = (y - x + \mu + \sigma^2\lambda)/\sigma$ we have

$$s(x|\mu, \sigma, n, \lambda) = \frac{\lambda^n \sigma^{n-1}}{\sqrt{2\pi}(n-1)!} e^{E_u(x|\lambda,\sigma,\mu)} \int_{\tilde{x}_u}^{+\infty} (\tau - \tilde{x}_u)^{n-1} e^{-\tau^2/2} d\tau,$$

where $\tilde{x}_u = -x/\sigma + (\mu + \sigma^2\lambda)/\sigma$. Now denote by

$$\mathrm{Hh}_n(x) = \frac{1}{n!}\int_x^{+\infty} (\tau - x)^n \, e^{-\tau^2/2} d\tau, \tag{27}$$

finally we have then

$$s(x|\mu, \sigma, n, \lambda) = \frac{\lambda^n \sigma^{n-1}}{\sqrt{2\pi}} e^{E_u(x|\lambda,\sigma,\mu)} \, \mathrm{Hh}_{n-1}(\tilde{x}_u), \quad n > 0. \tag{28}$$

A similar formula can be found in the Appendix of Kou [20] that relies on the integrals $\mathrm{Hh}(\tilde{x}_u)$ in terms of confluent hypergeometric functions, however these integrals can be expressed in terms of simple functions using the binomial theorem

$$\mathrm{Hh}_n(x) = \frac{1}{n!}\sum_{k=0}^{n} \binom{n}{k} (-x)^{n-k} \int_x^{+\infty} \tau^k e^{-\tau^2/2} d\tau = \sqrt{2\pi} \sum_{k=0}^{n} \frac{(-x)^{n-k}}{k!(n-k)!} J_k(x). \tag{29}$$

Now define

$$J_k(x) = \frac{1}{\sqrt{2\pi}} \int_x^{+\infty} y^k e^{-y^2/2} dy \qquad k \in \mathbb{N}, \tag{30}$$

and observe that

$$J_0(x) = \frac{1}{2}\,\mathrm{erfc}\left(\frac{x}{\sqrt{2}}\right), \tag{31}$$

with the change of variables $x^2/2 = t$ we can write $J_k(x)$ in terms of the upper and lower incomplete gamma functions denoted with $\Gamma(n, x)$ and $\gamma(n, x)$, respectively

$$J_k(x) = \begin{cases} \frac{2^{\frac{k-1}{2}}}{\sqrt{2\pi}} \Gamma\left(\frac{k+1}{2}, \frac{x^2}{2}\right), x \geq 0 \\ \frac{2^{\frac{k-1}{2}}}{\sqrt{2\pi}} \left[(-1)^k \gamma\left(\frac{k+1}{2}, \frac{x^2}{2}\right) + \Gamma\left(\frac{k+1}{2}\right)\right], x < 0 \end{cases}$$

Remark 1. *Equations (28) and (29) are significant for parameter estimation, although further exploration is necessary because a plain maximum log-likelihood estimation method may not be advisable because of the summation over terms within the logarithm of the likelihood function for $n \geq 2$ (see Equation (29)). However, alternative techniques such as the EM method or stochastic gradient descent might be more suitable (see Bishop [5]). The computational expense of these two methods is nevertheless lower compared to the Monte Carlo-based (MC) techniques commonly utilized for this purpose (see Laudage et al. [22]).*

We remark that the incomplete gamma functions above can be written in terms of exponential functions since they depend on an integer parameter[2].

In addition, we compute the following two integrals, that we will show, are a key component of the pricing formulas. The first one is given by proposition B.2 from Kou [20]

[2] In our numerical experiments we rely on the Python package *scipy*, see `https://docs.scipy.org/doc/scipy/reference/generated/scipy.special.gammainc.html` and `https://www.boost.org/doc/libs/1_61_0/libs/math/doc/html/math_toolkit/sf_gamma/igamma.html`

Proposition 4.1. *Define the integral*

$$I_n(c\,|\,\alpha,\beta,\delta) = \int_c^\infty e^{\alpha x}\,\mathrm{Hh}_n(\beta x-\delta)dx \tag{32}$$

then

1. $\beta > 0$ $\alpha \neq 0$, $\forall n \geq -1$

$$I_n(c\,|\,\alpha,\beta,\delta) = -\frac{e^{\alpha c}}{\alpha}\sum_{i=0}^{n}\left(\frac{\beta}{\alpha}\right)^{n-i}\mathrm{Hh}_i(\beta c-\delta)+\left(\frac{\beta}{\alpha}\right)^{n+1}\frac{\sqrt{2\pi}}{\beta}e^{\frac{\alpha\delta}{\beta}+\frac{\alpha^2}{2\beta^2}}\Phi\left(-\beta c+\delta+\frac{\alpha}{\beta}\right) \tag{33}$$

2. $\beta < 0$ $\alpha < 0$, $\forall n \geq -1$

$$I_n(c\,|\,\alpha,\beta,\delta) = -\frac{e^{\alpha c}}{\alpha}\sum_{i=0}^{n}\left(\frac{\beta}{\alpha}\right)^{n-i}\mathrm{Hh}_i(\beta c-\delta)-\left(\frac{\beta}{\alpha}\right)^{n+1}\frac{\sqrt{2\pi}}{\beta}e^{\frac{\alpha\delta}{\beta}+\frac{\alpha^2}{2\beta^2}}\Phi\left(\beta c-\delta-\frac{\alpha}{\beta}\right). \tag{34}$$

The second integral is given by the following proposition.

Proposition 4.2. *Define*

$$M_n(c\,|\,\alpha,\beta,\delta) = \int_c^\infty x\,e^{\alpha\,x}Hh_n(\beta x-\delta)dx, \quad n \geq -1.$$

Upon the cases of Proposition 4.1 for arbitrary constants α,β,c,δ, with $\alpha \neq 0$ the following recursion holds

$$M_n(c\,|\,\alpha,\beta,\delta) \;=\; \frac{1}{\alpha}\left[-c\,e^{\alpha c}Hh_n(\beta c-\delta) - I_n(c\,|\,\alpha,\beta,\delta) + \beta\,M_{n-1}(c,\alpha,\beta,\delta)\right] \tag{35}$$

The recursion is initialized taking $Hh_{-1}(x) = e^{-\frac{x^2}{2}}$

$$M_{-1}(c\,|\,\alpha,\beta,\delta) \;=\; \int_c^\infty x\,e^{\alpha\,x}e^{-\frac{(\beta x-\delta)^2}{2}}dx = \sqrt{2\pi}\,v\,e^{\frac{A}{2}}\left(\sigma^2\,g(c|\,m,v)+m\,\bar{\Phi}(c|\,m,v)\right)$$

with $A = (\alpha/\beta+\delta)^2-\delta^2, m = (\alpha+\beta\delta)/\beta^2, v = 1/|\beta|$

Proof. Integration by parts leads

$$\begin{aligned} M_n(c\,|\,\alpha,\beta,\delta) &= \frac{1}{\alpha}\,x\,e^{\alpha x}Hh_n(\beta x-\delta)\Big|_c^\infty \\ &- \frac{1}{\alpha}\int_c^\infty e^{\alpha x}\left[Hh_n(\beta x-\delta)-\beta x\,Hh_{n-1}(\beta x-\delta)\right]dx. \end{aligned}$$

Due to the results Kou [20] Section B.2, it turns out that the integral is convergent for $(\alpha,\beta) \notin \mathcal{D} = \{(\alpha,\beta),\alpha > 0,\beta \leq 0\}$ which concludes the proof. $\square$

4.2 Convolution of a Gaussian and bilateral Erlang law

Consider $X + Y$ where $Y \sim b\,\mathcal{E}_n(\lambda)$. Its *pdf* is obtained once again via convolution.

$$\begin{aligned} b(x|\mu,\sigma,n,\lambda) = & \frac{\lambda^n}{\sqrt{2\pi}\sigma(n-1)!}\sum_{k=0}^{n-1}\frac{\omega_{k,n}}{(n-1-k)!2^n} \\ & \times \int_{\mathbb{R}} |y|^{n-k-1}\exp\left(-\frac{(x-y-\mu)^2+2\sigma^2\lambda|y|}{2\sigma^2}\right)dy. \end{aligned}$$

If we split the domain of integration, we obtain that

$$\begin{aligned} b(x|\mu,\sigma,n,\lambda) = & \frac{\lambda^n}{\sqrt{2\pi}\sigma(n-1)!}\sum_{k=0}^{n-1}\frac{\omega_{k,n}}{(n-1-k)!2^n} \\ & \times \left[\int_{\mathbb{R}^-}(-y)^{n-k-1}\exp\left(-\frac{(x-y-\mu)^2-2\sigma^2\lambda y}{2\sigma^2}\right)dy\right. \\ & \left.+\int_{\mathbb{R}^+}(y)^{n-k-1}\exp\left(-\frac{(x-y-\mu)^2+2\sigma^2\lambda y}{2\sigma^2}\right)dy\right] \end{aligned}$$

After an intermediate change of variable $\tilde{y} = -y$, and then renaming back $\tilde{y} = y$, $b(x|\mu,\sigma,n,\lambda)$ can be rewritten as

$$\begin{aligned} b(x|\mu,\sigma,n,\lambda) = & \frac{\lambda^n}{\sqrt{2\pi}\sigma(n-1)!}\sum_{k=0}^{n-1}\frac{\omega_{k,n}}{(n-1-k)!2^n} \\ & \times \left[\int_{\mathbb{R}^+}y^{n-k-1}\exp\left(-\frac{(x+y-\mu)^2+2\sigma^2\lambda y}{2\sigma^2}\right)dy+\right. \\ & \left.+\int_{\mathbb{R}^+}y^{n-k-1}\exp\left(-\frac{(x-y-\mu)^2+2\sigma^2\lambda y}{2\sigma^2}\right)dy.\right] \end{aligned}$$

Now, focus on the argument of the exponential of the first integral

$$A_d(x,y|\lambda,\sigma,\mu) = (x+y-\mu)^2 + 2\sigma^2\lambda y,$$

and on the quantity

$$(x+y-\mu+\sigma^2\lambda)^2 = (x+y-\mu)^2 + \sigma^4\lambda^2 + 2\sigma^2\lambda(x+y-\mu).$$

Therefore, we have

$$A_d(x,y|\lambda,\sigma,\mu) = (x+y-\mu+\sigma^2\lambda)^2 - \sigma^4\lambda^2 - 2\sigma^2\lambda(x-\mu).$$

Using the results in Section 4.1 it results

$$\begin{aligned} b(x|\mu,\sigma,n,\lambda) = & \frac{\lambda^n}{\sqrt{2\pi}2^n(n-1)!}\sum_{k=0}^{n-1}\sigma^{n-k-1}\omega_{k,n}\left(e^{E_d(x|\,\lambda,\sigma,\mu)}\mathrm{Hh}_{n-k-1}(\tilde{x}_d)+\right. \\ & \left. e^{E_u(x|\,\lambda,\sigma,\mu)}\mathrm{Hh}_{n-k-1}(\tilde{x}_u)\right), \end{aligned} \tag{36}$$

where $E_d(x|\,\lambda,\sigma,\mu) = \lambda x + \frac{1}{2}(\sigma^2\lambda^2 - 2\mu\lambda)$ and $\tilde{x}_d = \frac{x}{\sigma} + \frac{\sigma^2\lambda-\mu}{\sigma}$.

4.3 Convolution of a Gaussian and Erlang-difference law

Consider $W = X + Y_u - Y_d$ where $Y_u \sim \mathcal{E}_n(\lambda_u)$ and $Y_d \sim \mathcal{E}_m(\lambda_d)$ where the *pdf* of $Y_u - Y_d$ is given by Equation (11). With the same logic of the previous subsections the *pdf* of W is obtained by convolution

$$
\begin{aligned}
d(x|\mu,\sigma,n,m,\lambda_u,\lambda_d) =& \int_{\mathbb{R}} g_{\mu,\sigma}(x-y) f(y|n,m,\lambda_u,\lambda_d) dy = \\
=& \frac{\lambda_d^m}{\sqrt{2\pi}\sigma(n-1)!}\left(\frac{\lambda_u}{\lambda_d+\lambda_u}\right)^n \sum_{\ell=0}^{m-1} \frac{\alpha_{\ell,n}}{(m-\ell-1)!} \\
& \times \int_{\mathbb{R}^-} (-y)^{m-\ell-1} e^{\lambda_d y} e^{-\frac{(x-y-\mu)^2}{2\sigma^2}} dy + \\
& + \frac{\lambda_u^n}{\sqrt{2\pi}\sigma(m-1)!}\left(\frac{\lambda_d}{\lambda_d+\lambda_u}\right)^m \sum_{k=0}^{n-1} \frac{\alpha_{k,m}}{(n-k-1)!} \\
& \times \int_{\mathbb{R}^+} (y)^{n-k-1} e^{-\lambda_u y} e^{-\frac{(x-y-\mu)^2}{2\sigma^2}} dy.
\end{aligned}
$$

Adopting the same change of variable of Section 4.2 gives

$$
\begin{aligned}
d(x|\mu,\sigma,n,m,\lambda_u,\lambda_d) =& \frac{\lambda_d^m}{(n-1)!}\left(\frac{\lambda_u}{\lambda_d+\lambda_u}\right)^n \frac{1}{\sqrt{2\pi}\sigma} \sum_{\ell=0}^{m-1} \frac{\alpha_{\ell,n}}{(m-\ell-1)!} \\
& \times \int_{\mathbb{R}^+} y^{m-\ell-1} e^{-\frac{A_d(x,y|\lambda_d,\sigma,\mu)}{2\sigma^2}} dy + \\
& + \frac{\lambda_u^n}{(m-1)!}\left(\frac{\lambda_d}{\lambda_u+\lambda_d}\right)^m \frac{1}{\sqrt{2\pi}\sigma} \sum_{k=0}^{n-1} \frac{\alpha_{k,m}}{(n-k-1)!} \\
& \times \int_{\mathbb{R}^+} y^{n-k-1} e^{-\frac{A_u(x,y|\lambda_u,\sigma,\mu)}{2\sigma^2}} dy.
\end{aligned}
$$

Once again, the density can be written in terms of the $\mathrm{Hh}_k(\cdot)$ functions as follows

$$
\begin{aligned}
d(x|\mu,\sigma,n,m,\lambda_u,\lambda_d) =& \frac{1}{\sqrt{2\pi}} \times \\
& \Bigg[\frac{\lambda_d^m}{(n-1)!}\left(\frac{\lambda_u}{\lambda_d+\lambda_u}\right)^n \sum_{\ell=0}^{m-1} \alpha_{\ell,n} \sigma^{m-\ell-1} e^{E_d(x|\lambda_d,\sigma,\mu)} \mathrm{Hh}_{m-\ell-1}(\tilde{x}_d) + \\
& + \frac{\lambda_u^n}{(m-1)!}\left(\frac{\lambda_d}{\lambda_d+\lambda_u}\right)^m \sum_{k=0}^{n-1} \alpha_{k,m} \sigma^{n-k-1} e^{E_u(x|\lambda_u,\sigma,\mu)} \mathrm{Hh}_{n-k-1}(\tilde{x}_u) \Bigg].
\end{aligned} \tag{37}
$$

4.4 Convolution of Gaussians and the sum of i.i.d. double exponential Laws

Finally, consider $X+Y$ where $X \sim \mathcal{N}(\mu, \sigma^2)$ and Y is the sum of n i.i.d. double exponential *rv*'s whose law can be seen as the mixture of a positive exponential $Y_u \sim \mathcal{E}_1(\lambda_u)$ and a negative exponential $-Y_d \sim \mathcal{E}_1(\lambda_d)$ law with mixture parameters p and $q = 1-p$ with *pdf* defined in Equation (13) and with the following *chf*

$$\varphi(u|\lambda_u, \lambda_d, p) = p\frac{\lambda_u}{\lambda_u - iu} + (1-p)\frac{\lambda_d}{\lambda_d + iu}. \tag{38}$$

Therefore the *chf* of the sum of n i.i.d. *rv*'s is

$$\begin{aligned}\varphi(u|, n, p, \lambda_u, \lambda_d) &= \left(p\frac{\lambda_u}{\lambda_u - iu} + (1-p)\frac{\lambda_d}{\lambda_d + iu}\right)^n \\ &= \sum_{k=0}^{n} \binom{n}{k} p^k (1-p)^{n-k} \left(\frac{\lambda_u}{\lambda_u - iu}\right)^k \left(\frac{\lambda_d}{\lambda_d + iu}\right)^{n-k}\end{aligned} \tag{39}$$

and its law has *pdf*

$$\ell(x|n, p, \lambda_u, \lambda_d) = \sum_{k=0}^{n} \binom{n}{k} p^k (1-p)^{n-k} f(x|n, n-k, \lambda_u, \lambda_d) \tag{40}$$

where $f(\cdot|n, p, \lambda_u, \lambda_d)$ is defined in Equation (11). We can then conclude that the sum of n i.i.d. double exponential *rv*'s can also be seen as a mixture, now of the difference of Erlang laws with mixture parameters given by a binomial distribution $\mathfrak{B}(p, n)$. This result can be very useful to draw a *rv* from such a law indeed, the simulation consists of simply generating a binomial *rv* $B \sim \mathfrak{B}(n, p)$ and given B generating the difference of Erlang $Y_u - Y_d$ where $Y_u \sim \mathcal{E}_B(\lambda_u)$ and $Y_d \sim \mathcal{E}_{n-B}(\lambda_d)$.

Finally, we can derive the *pdf* of $X+Y$

$$k(x|\mu, \sigma, n, p, \lambda_u, \lambda_d) = \sum_{k=0}^{n} \binom{n}{k} p^k (1-p)^{n-k} \int_{\mathbb{R}} g(x-y|\mu, \sigma)\, f(y|n, n-k, \lambda_u, \lambda_d) dy$$

Therefore, using the result of Subsection 4.3 one can rewrite the *pdf* above as

$$k(x|\mu, \sigma, n, p, \lambda_u, \lambda_d) = \sum_{k=0}^{n} \binom{n}{k} p^k (1-p)^{n-k}\, d(x|\mu, \sigma, n, n-k, \lambda_u, \lambda_d). \tag{41}$$

5 Closed-formulas for standard options

In this section, we derive closed formulas for European call and put options based on the models described by equation (17) and (18). To this end, we only need to find the expressions for Ψ_1, Ψ_2 and Ξ for different types of jump distribution.

5.1 Exponential distributed jumps

Based on the results of Sabino and Cufaro Petroni [33] if $J_n \sim \mathcal{E}_1(\lambda_J)$ then $\Sigma_J(T, T_1, T_2)\, Z$ is an Erlang-Pólya mixture $\mathcal{E}_P(\lambda)$ with $\lambda = \lambda_J/(a\, \Sigma_J(T, T_1, T_2))$, $a = e^{-k_J T}$ and P is distributed according to a Pólya $\overline{\mathfrak{B}}(\alpha, p)$ with $\alpha = \lambda_P/k_J$ and $p = 1 - a$. Hence, the law of E given $\Sigma_J(T, T_1, T_2)\, Z$ is that of the sum of a Gaussian with parameters $\mu = 0$ and $\sigma^2 = \sigma^2(T, T_1, T_2) = \Sigma_D^2(T, T_1, T_2)\left(\frac{1-e^{2k_D T}}{2k_D}\right) + \Gamma_2(T_1, T_2)^2$ and an Erlang with parameter λ. The *chf* of Z is then

$$\phi_Z(u) = \left(\frac{\lambda_J - iau}{\lambda_J - iu}\right)^{\alpha}. \tag{42}$$

Of course, from the equation above one can get the *cgf* and obtain $h(T, T_1, T_2)$ from Equation (24).

Then, defining $p_n = \boldsymbol{P}\{P = n\}$, we have

$$\Psi_1(c) = \sum_{n=0}^{\infty} p_n \int_c^{\infty} e^x\, s(x|\mu, \sigma, n, \lambda)\, dx = \sum_{n=0}^{\infty} p_n \Psi_{1,n}(c), \tag{43}$$

$$\Psi_2(c) = \sum_{n=0}^{\infty} p_n \int_c^{\infty} s(x|\mu, \sigma, n, \lambda)\, dx = \sum_{n=0}^{\infty} p_n \Psi_{2,n}(c), \tag{44}$$

$$\Xi(c) = \sum_{n=0}^{\infty} p_n \int_c^{\infty} x\, s(x|\mu, \sigma, n, \lambda)\, dx = \sum_{n=0}^{\infty} p_n \Xi_n(c). \tag{45}$$

Consequently, from Propositions 4.1 and 4.2 it results

$$\Psi_{1,n}(c) = \frac{\lambda^n \sigma^{n-1}}{\sqrt{2\pi}} e^{\frac{\sigma^2\lambda^2 + 2\mu\lambda}{2}} I_{n-1}\left(c\,|\,1 - \lambda, -\frac{1}{\sigma}, -\frac{\mu + \sigma^2\lambda}{\sigma}\right) \tag{46}$$

$$\Psi_{2,n}(c) = \frac{\lambda^n \sigma^{n-1}}{\sqrt{2\pi}} e^{\frac{\sigma^2\lambda^2 + 2\mu\lambda}{2}} I_{n-1}\left(c\,|\, - \lambda, -\frac{1}{\sigma}, -\frac{\mu + \sigma^2\lambda}{\sigma}\right) \tag{47}$$

$$\Xi_n(c) = \frac{\lambda^n \sigma^{n-1}}{\sqrt{2\pi}} e^{\frac{\sigma^2\lambda^2 + 2\mu\lambda}{2}} M_{n-1}\left(c\,|\, - \lambda, -\frac{1}{\sigma}, -\frac{\mu + \sigma^2\lambda}{\sigma}\right) \tag{48}$$

The calculation of the terms $I_n, n \geq -1$ is covered by the case 2 of Propositions 4.1.

5.2 Bilateral exponential jumps

In this section we assume that J_n is distributed according to a bilateral exponential law, also known as Laplace distribution. Once again based on the results of Sabino and Cufaro Petroni [34], $\Sigma_J(T, T_1, T_2)Z$ is now a bilateral Erlang-Pólya mixture $b\,\mathcal{E}_{\hat{P}}(\lambda)$ with $\lambda = \lambda_J/(a\, \Sigma_J(T, T_1, T_2))$, $a = e^{-k_J T}$ and $\hat{P}$ is distributed according to a Pólya $\overline{\mathfrak{B}}(\alpha, p)$ with $\alpha = \lambda_P/2k_J$ and $p = 1 - a^2$. Now, the law of E given

$\Sigma_J(T, T_1, T_2)\, Z$ is that of the sum of a Gaussian with parameters $\mu = 0$ and $\sigma^2 = \sigma^2(T, T_1, T_2)$ and an bilateral Erlang with parameter λ. The *chf* of Z is then

$$\phi_Z(u) = \left(\frac{\lambda_J^2 + a^2 u^2}{\lambda_J^2 + u^2} \right)^{\alpha}. \tag{49}$$

$h(T, T_1, T_2)$ is accordingly obtained from Equation (24).

In contrast to Equations (43)-(45) for the exponential jumps, one needs to take a different $p_n = \boldsymbol{P}\left\{\hat{P} = n\right\}$ and the *pdf* $b(\cdot|\mu, \sigma, n, \lambda)$ defined in Equation (36). Accordingly, from Propositions 4.1 and 4.2 we have for $n > 0$

$$\begin{aligned}
\Psi_{1,n}(c) =& \frac{\lambda^n}{\sqrt{2\pi} 2^n (n-1)!} \sum_{k=0}^{n-1} \omega_{k,n} \sigma^{n-k-1} \left[e^{\frac{1}{2}(\sigma^2\lambda^2 - 2\mu\lambda)} I_{n-k-1}\left(c, 1+\lambda, \frac{1}{\sigma}, \frac{\mu - \sigma^2\lambda}{\sigma}\right) \right. \\
&\left. + e^{\frac{1}{2}(\sigma^2\lambda^2 + 2\mu\lambda)} I_{n-k-1}\left(c, 1-\lambda, -\frac{1}{\sigma}, -\frac{\mu + \sigma^2\lambda}{\sigma}\right) \right], \qquad (50) \\
\Psi_{2,n}(c) =& \frac{\lambda^n}{\sqrt{2\pi} 2^n (n-1)!} \sum_{k=0}^{n-1} \omega_{k,n} \sigma^{n-k-1} \left[e^{\frac{1}{2}(\sigma^2\lambda^2 - 2\mu\lambda)} I_{n-k-1}\left(c, \lambda, \frac{1}{\sigma}, \frac{\mu - \sigma^2\lambda}{\sigma}\right) \right. \\
&\left. + e^{\frac{1}{2}(\sigma^2\lambda^2 + 2\mu\lambda)} I_{n-k-1}\left(c, -\lambda, -\frac{1}{\sigma}, -\frac{\mu + \sigma^2\lambda}{\sigma}\right) \right]. \qquad (51) \\
\Xi_n(c) =& \frac{\lambda^n}{\sqrt{2\pi} 2^n (n-1)!} \sum_{k=0}^{n-1} \omega_{k,n} \sigma^{n-k-1} \left[e^{\frac{1}{2}(\sigma^2\lambda^2 - 2\mu\lambda)} M_{n-k-1}\left(c, \lambda, \frac{1}{\sigma}, \frac{\mu - \sigma^2\lambda}{\sigma}\right) \right. \\
&\left. + e^{\frac{1}{2}(\sigma^2\lambda^2 + 2\mu\lambda)} M_{n-k-1}\left(c, -\lambda, -\frac{1}{\sigma}, -\frac{\mu + \sigma^2\lambda}{\sigma}\right) \right]. \qquad (52)
\end{aligned}$$

Case 1 of Proposition 4.1 covers the first and third occurrence of the integral $I_{n-k-1}(\cdot)$ and the first of $M_{n-k-1}(\cdot)$, whereas case 2 covers the remaining ones.

5.3 Difference of two independent jump processes

In order to separate downward and upward jumps occurrences it may be useful to introduce two independent jump processes

$$Z_r \stackrel{d}{=} \sum_{n_r=0}^{N_r(t)} e^{-k_{r,J}(t-\tau_{r,n_r}})J_{r,n_r}, \quad r \in \{u, d\}. \tag{53}$$

and also two deterministic functions $\Sigma_r(t, T_1, T_2), r \in \{u, d\}$ with parameters γ_r and k_r, $r \in \{u, d\}$. The Poisson process $N_r(t)$ has intensity $\lambda_{r,P}$ and $J_r \sim \mathcal{E}_1(\lambda_{r,J})$.

Hence,

$$
\begin{aligned}
\Psi_1(c) =& \sum_{n=0}^{\infty}\sum_{m=0}^{\infty} u_n d_m \int_c^{\infty} e^x\, d(x|\mu,\sigma,n,m,\lambda_u,\lambda_d)(x)\,dx = \sum_{n=0}^{\infty}\sum_{m=0}^{\infty} u_n d_m \Psi_{1,n,m}(c),\\
\Psi_2(c) =& \sum_{n=0}^{\infty}\sum_{m=0}^{\infty} u_n d_m \int_c^{\infty} d(x|\mu,\sigma,n,m,\lambda_u,\lambda_d)\,dx = \sum_{n=0}^{\infty}\sum_{m=0}^{\infty} u_n d_m \Psi_{2,n,m}(c)\\
\Xi(c) =& \sum_{n=0}^{\infty}\sum_{m=0}^{\infty} u_n d_m \int_c^{\infty} x\, d(x|\mu,\sigma,n,m,\lambda_u,\lambda_d)\,dx = \sum_{n=0}^{\infty}\sum_{m=0}^{\infty} u_n d_m \Xi_{n,m}(c)
\end{aligned}
$$

where the law of E given $\Sigma_u(T,T_1,T_2)Z_u(T)$ and $\Sigma_d(T,T_1,T_2)Z_d$ is the convolution of a Gaussian law with parameters $\mu = 0$ and $\sigma^2 = \sigma^2(T,T_1,T_2)$ and the difference of two independent Erlang distributions with upward and downward parameters $\lambda_r = \lambda_{r,J}/(a_r\,\Sigma_d(T,T_1,T_2))$, $a_u = e^{-k_{u,J}T}, d \in \{u,d\}$, respectively. Moreover, $u_n = \boldsymbol{P}\{P_u = n\}$ with $P_u \sim \overline{\mathfrak{B}}(\lambda_{u,P}/k_{u,J}, 1-a_u)$, $d_n = \boldsymbol{P}\{P_d = m\}$ and $P_d \sim \overline{\mathfrak{B}}(\lambda_{d,P}/k_{d,J}, 1-a_d)$, $a_d = e^{-k_{d,J}T}$. In addition, the *chf* of Z is

$$
\phi_Z(v) = \left(\frac{\lambda_{u,J} - i a_u v}{\lambda_{u,J} - iv}\right)^{\frac{\lambda_{u,P}}{k_{u,J}}} \left(\frac{\lambda_{d,J} + i a_d v}{\lambda_{d,J} + iv}\right)^{\frac{\lambda_{d,P}}{k_{d,J}}}. \tag{54}
$$

Once again, $h(T,T_1,T_2)$ is obtained from the Equation above and (24).

Analogously to the cases above, from Propositions 4.1 and 4.2 it turns out

$$
\begin{aligned}
\Psi_{1,n,m}(c) =& \frac{1}{\sqrt{2\pi}} \left\{ \frac{\lambda_d^m}{(n-1)!} \left(\frac{\lambda_u}{\lambda_d+\lambda_u}\right)^n \times \right.\\
& \times \sum_{k=0}^{m-1} \alpha_{k,n,m} \sigma^{m-k-1} e^{\frac{1}{2}(\sigma^2\lambda_d^2 - 2\mu\lambda_d)} I_{m-k-1}\left(c, 1+\lambda_d, \frac{1}{\sigma}, \frac{\mu - \sigma^2\lambda_d}{\sigma}\right) + \\
& + \frac{\lambda_u^n}{(m-1)!} \left(\frac{\lambda_d}{\lambda_u+\lambda_d}\right)^m \times \\
& \left. \times \sum_{\ell=0}^{n-1} \alpha_{\ell,n,m} \sigma^{n-\ell-1} e^{\frac{1}{2}(\sigma^2\lambda_u^2 + 2\mu\lambda_u)} I_{n-\ell-1}\left(c, 1-\lambda_u, -\frac{1}{\sigma}, -\frac{\mu+\sigma^2\lambda_u}{\sigma}\right) \right\},
\end{aligned} \tag{55}
$$

$$
\begin{aligned}
\Psi_{2,n,m}(c) =& \frac{1}{\sqrt{2\pi}} \left\{ \frac{\lambda_d^m}{(n-1)!} \left(\frac{\lambda_u}{\lambda_d+\lambda_u} \right)^n \times \right. \\
& \times \sum_{k=0}^{m-1} \alpha_{k,n,m} \sigma^{m-k-1} e^{\frac{1}{2}(\sigma^2\lambda_d^2 - 2\mu\lambda_d)} I_{m-k-1}\left(c, \lambda_d, \frac{1}{\sigma}, \frac{\mu-\sigma^2\lambda_d}{\sigma} \right) + \\
& + \frac{\lambda_u^n}{(m-1)!} \left(\frac{\lambda_d}{\lambda_u+\lambda_d} \right)^m \times \\
& \left. \times \sum_{\ell=0}^{n-1} \alpha_{\ell,n,m} \sigma^{n-\ell-1} e^{\frac{1}{2}(\sigma^2\lambda_u^2 + 2\mu\lambda_u)} I_{n-\ell-1}\left(c, -\lambda_u, -\frac{1}{\sigma}, -\frac{\mu+\sigma^2\lambda_u}{\sigma} \right) \right\}. \quad (56)
\end{aligned}
$$

(57)

$$
\begin{aligned}
\Xi_n(c,m) =& \frac{1}{\sqrt{2\pi}} \left\{ \frac{\lambda_d^m}{(n-1)!} \left(\frac{\lambda_u}{\lambda_d+\lambda_u} \right)^n \times \right. \\
& \times \sum_{k=0}^{m-1} \alpha_{k,n,m} \sigma^{m-k-1} e^{\frac{1}{2}(\sigma^2\lambda_d^2 - 2\mu\lambda_d)} M_{m-k-1}\left(c, \lambda_d, \frac{1}{\sigma}, \frac{\mu-\sigma^2\lambda_d}{\sigma} \right) + \\
& + \frac{\lambda_u^n}{(m-1)!} \left(\frac{\lambda_d}{\lambda_u+\lambda_d} \right)^m \times \\
& \left. \times \sum_{\ell=0}^{n-1} \alpha_{\ell,n,m} \sigma^{n-\ell-1} e^{\frac{1}{2}(\sigma^2\lambda_u^2 + 2\mu\lambda_u)} M_{n-\ell-1}\left(c, -\lambda_u, -\frac{1}{\sigma}, -\frac{\mu+\sigma^2\lambda_u}{\sigma} \right) \right\},
\end{aligned}
\tag{58}
$$

5.4 Double exponential jumps

Finally, we consider the closest setting to the original Kou model, namely when J_n follows a double exponential law. Following Cont and Tankov [7] and Kluge [19] it results that the *chf* of Z, is given by

$$
\log \varphi_Z(v) = \lambda_P \int_0^T \left(\varphi_J(ve^{-k_J s}) - 1 \right) ds,
$$

where $\varphi_J(v)$ is the *chf* of the double exponential law with *pdf* in Equation 13. Denoting $\varphi_u(v) = \lambda_{u,J}/(\lambda_u - iv)$ and $\varphi_d(v) = \lambda_{d,J}/(\lambda_d + iv)$ we have

$$
\log \varphi_Z(v) = p\lambda_P \int_0^T \left(\varphi_u(ve^{-k_J s}) - 1 \right) ds + (1-p)\lambda_P \int_0^t \left(\varphi_d(ve^{-k_J s}) - 1 \right) ds,
$$

hence

$$
\varphi_Z(v) = \left(\frac{\lambda_{u,J} - iue^{-k_J T}}{\lambda_{u,J} - iv} \right)^{\frac{p\lambda_P}{k_J}} \left(\frac{\lambda_{d,J} + ive^{-k_J T}}{\lambda_{d,J} + iv} \right)^{\frac{(1-p)\lambda_P}{k_J}}, \tag{59}
$$

and $h(T, T_1, T_2)$ is obtained as usual from Equation (24). It means that we can write $\Sigma_J(T, T_1, T_2) Z \stackrel{d}{=} Z_u - Z_d$ where $Z_r \sim \mathcal{E}_{P_r}(\lambda_r)$, $\lambda_r = \lambda_{r,J}/(a\Sigma_J(T, T_1, T_2))$, $r \in \{u, d\}$

with $a = e^{-k_J T}$, $P_u \sim \overline{\mathfrak{B}}(p\lambda_P/k, 1-a)$ and $P_d \sim \overline{\mathfrak{B}}((1-p)\lambda_P/k, 1-a)$. Consequently, the functions Ψ_1, Ψ_2, Xi are still given by Equations (55)-(58), respectively taking $u_n = \boldsymbol{P}\{P_u = n\}$ and $d_m = \boldsymbol{P}\{P_d = m\}$.

6 Numerical experiments

This section delves into a comprehensive analysis of the performance and efficacy of the option formulas derived in the preceding sections. All numerical analyses within this study were executed using Python on a 64-bit Intel Core i5-6300U CPU with 8 GB of memory. The evaluation of algorithmic performance primarily relies on the difference between the option price obtained with 10^5 repetitions with Monte Carlo (MC) and the closed-formulas (see Sabino and Cufaro Petroni [33, 34] for methodological details on the MC algorithm).

This analysis encompasses three jump process configurations: pure upward jumps, bilateral jumps, and double exponential jumps. Additionally, both the NOA and exponential settings are considered. We consider the following set of parameters:

- $F(0, T_1, T_2) = 100$.
- $T_0 \in \{0.25, 0.5, 0.75, 1\}$, $T_1 = 1$, $T_2 = 2$.
- $K \in \{80, 85, 90, 95, 100, 105, 110, 115, 120\}$.
- $\mu = 0$, $\lambda_P = 5$, $\lambda_u = 2$, $\lambda_d = 0.7$, $p = 0.4$.
- $k_D = 0.8$, $k_J = 0.7$.
- $\gamma_D = 0.5$, $\gamma_J = 0.3$, $\gamma_2 = 0.18$.

Table 1: Jumps up, exponential setting, numbers are multiplied by a factor 10^4.

	MC-Analytical	err	MC-Analytical	err	MC-Analytical	err	MC-Analytical	err
$K\backslash T_0$	0.25		0.5		0.75		1	
80	9.5	10	12	14	15	18	20	21
85	8.6	9.6	13	14	17	17	20	21
90	0.9	9.3	13	13	15	17	20	21
95	2.4	8.9	12	13	13	17	20	20
100	7.7	8.5	12	12	16	16	19	20
105	6.5	8.1	11	12	13	16	20	20
110	7.2	7.8	11	12	14	16	19	19
115	6.3	7.5	11	11	13	15	19	19
120	6.2	7.2	5	11	13	15	18	19

The outcomes of the numerical experiments are presented in Tables 1- 6 with changing strike and maturity, featuring varying strike prices and maturities. In addition to the Monte Carlo (MC)-based results, we have included the Root Mean Square

Table 2: Jumps up, NOA setting, numbers are multiplied by a factor 10^4.

	MC-Analytical	err	MC-Analytical	err	MC-Analytical	err	MC-Analytical	err
$K\backslash T_0$	0.25		0.5		0.75		1	
80	0.56	0.87	0.96	1.08	0.94	1.26	0.94	1.43
85	0.63	0.83	0.95	1.04	0.98	1.21	0.98	1.38
90	0.66	0.79	0.93	0.99	1.03	1.16	1.03	1.34
95	0.68	0.74	0.90	0.94	1.10	1.12	1.10	1.29
100	0.70	0.70	0.87	0.90	1.05	1.07	1.15	1.24
105	0.61	0.65	0.79	0.85	0.99	1.02	1.19	1.19
110	0.59	0.60	0.78	0.80	0.92	0.97	1.08	1.14
115	0.66	0.56	0.71	0.75	0.88	0.92	1.04	1.09
120	0.59	0.51	0.71	0.71	0.83	0.88	0.98	1.04

Table 3: Bilateral jumps, exponential setting, numbers are multiplied by a factor 10^3.

	MC-Analytical	err	MC-Analytical	err	MC-Analytical	err	MC-Analytical	err
$K\backslash T_0$	0.25		0.5		0.75		1	
80	1.2	1.4	1.7	1.9	2.4	2.5	2.8	3.2
85	1.2	1.4	1.6	1.9	2.4	2.5	2.9	3.1
90	1.2	1.3	1.5	1.9	2.2	2.4	3.0	3.1
95	1.3	1.3	1.4	1.8	2.3	2.4	2.9	3.1
100	1.2	1.3	1.4	1.8	2.2	2.4	9.9	3.0
105	1.2	1.2	1.4	1.7	2.1	2.4	3.0	3.0
110	1.1	1.2	1.6	1.7	2.3	2.3	2.7	3.0
115	1.0	1.1	1.4	1.7	2.2	2.3	2.6	3.0
120	1.0	1.1	1.6	1.6	2.0	2.3	2.6	2.9

Table 4: Bilateral jumps, NOA setting, numbers are multiplied by a factor 10^3.

	MC-Analytical	err	MC-Analytical	err	MC-Analytical	err	MC-Analytical	err
$K\backslash T_0$	0.25		0.5		0.75		1	
80	1.16	1.18	1.37	1.46	1.45	1.66	1.74	1.85
85	0.84	1.14	1.36	1.41	1.35	1.61	1.63	1.80
90	0.70	1.09	1.35	1.37	1.26	1.56	1.51	1.75
95	0.98	1.04	1.24	1.32	1.41	1.51	1.41	1.70
100	0.84	0.99	1.21	1.27	1.30	1.46	1.30	1.65
105	0.70	0.94	1.18	1.22	1.21	1.41	1.21	1.60
110	0.82	0.89	1.09	1.17	1.11	1.36	1.11	1.55
115	0.65	0.85	1.01	1.12	1.03	1.31	1.03	1.50
120	0.49	0.80	0.93	1.07	0.94	1.26	0.94	1.45

Table 5: Double exponential jumps, exponential setting, numbers are multiplied by a factor 10^3.

	MC-Analytical	err	MC-Analytical	err	MC-Analytical	err	MC-Analytical	err
$K \backslash T_0$	0.25		0.5		0.75		1	
80	0.97	1.35	1.17	1.44	4.54	5.16	1.94	2.43
85	1.23	1.61	1.22	1.39	4.59	5.05	2.02	2.39
90	1.41	1.54	1.22	1.35	4.61	4.94	2.08	2.35
95	1.59	1.76	1.29	1.31	4.62	4.84	2.14	2.32
100	1.63	1.68	1.31	1.69	4.62	4.73	2.22	2.28
105	1.66	1.73	1.35	1.64	4.61	4.92	2.04	2.24
110	1.65	1.77	1.38	1.58	4.61	4.81	2.08	2.21
115	1.60	1.68	1.49	1.53	4.61	4.70	2.11	2.17
120	1.52	1.59	1.46	1.48	4.59	4.60	2.13	2.14

Table 6: Double exponential jumps, NOA setting, numbers are multiplied by a factor 10^3.

	MC-Analytical	err	MC-Analytical	err	MC-Analytical	err	MC-Analytical	err
$K \backslash T_0$	0.25		0.5		0.75		1	
80	0.81	1.06	1.04	1.43	1173.69	1.75	1.87	2.05
85	0.85	1.02	1.21	1.38	1.30	1.70	1.86	2.00
90	0.40	0.97	1.22	1.33	1.28	1.65	1.83	1.95
95	0.66	0.92	1.19	1.28	1.26	1.60	1.79	1.90
100	0.69	0.87	1.13	1.23	1.14	1.55	1.79	1.85
105	0.75	0.82	1.15	1.18	1.04	1.50	1.79	1.79
110	0.71	0.77	1.11	1.12	0.97	1.44	1.68	1.74
115	0.69	0.72	1.06	1.07	0.95	1.39	1.64	1.69
120	0.66	0.67	0.99	1.02	0.93	1.33	1.61	1.63

Table 7: Computational times in seconds and ratios.

	Up		Bilateral		Double Exponential	
	Exponential	NOA	Exponential	NOA	Exponential	NOA
Analytical	0.75	0.75	0.93	0.94	1.21	1.22
MC	2.36	2.33	3.94	3.89	6.46	6.37
Ratio	3.14	3.10	4.23	4.14	5.34	5.22

Error (RMSE) to facilitate a more precise comparison with the analytically-based method. The tables illustrate that the option values derived from the analytical formulas align well with the MC results. Notably, none of the values exceeds three times the MC error, signifying a high level of compatibility between the methods. It is important to emphasize that the closed-formulas employed in this study do not rely on any numerical integration, not even the computation of the incomplete gamma function. The sole approximation utilized involves truncating the infinite series. Specifically, we included $n+1$ terms up $\boldsymbol{P}\{P>n\}=10^{-4}$ or $\boldsymbol{P}\{P_r>n\}=10^{-4}$, depending on the setting with $r\in\{u,d\}$. We then assigned the $n+1$th probability term the value 10^{-4} to ensure that the sum of all probability terms is normalized to 1. For instance, for the upward jump case, 20 terms were sufficient. Moreover, the numerical experiments with the analytical method seem to exhibit larger computation times. In contrast, the implementation of the simulation method relies on a fully Python-optimized solution in *scipy* written in low-level languages like Fortran, C and C++ which provides easy tools for array computing.

Consequently, based on the consistency of results and the methodology employed, we can confidently assert the accuracy and reliability of the closed-formula implementation.

7 Empirical study

In this section, we instantiate one of the models derived in the preceding sections and apply it to real market data. Specifically, we utilize the Kou model as introduced in Section 3, wherein the jump component is the summation of *rv*'s following a double exponential distribution. Within this conceptual framework, $F(t,T_1,T_2)$ are as in Equation (17)

$$\begin{aligned}\log F(t,T_1,T_2) =& \log F(0,T_1,T_2)+h(t,T_1,T_2)+\Gamma_2(T_1,T_2)W_2(t)\\ &+\Sigma_D(t,T_1,T_2)Z_D(t)+\Sigma_J(t,T_1,T_2)Z_J(t)\end{aligned}$$

and in Equation (19)

$$\begin{aligned}F(t,T_1,T_2) =& F(0,T_1,T_2)(1+h(t,T_1,T_2)+\Gamma_2(T_1,T_2)W_2(t)\\ &+\Sigma_D(t,T_1,T_2)Z_D(t)+\Sigma_J(t,T_1,T_2)Z_J(t))\end{aligned}$$

where J_n are *iid* *rv*'s with double exponential law with parameters λ_u,λ_p and p. In the subsequent section, we provide a succinct overview of numerical methods based on the an explicit form of the characteristic function. These methods are computationally efficient and hold significance for calibrating the model using real market data. Readers seeking detailed descriptions of particular methods may consult references such as Carr and Madan Carr [6], Fang and Oosterlee [11], and Lewis [23]. For a comprehensive overview, Schmelzle [36] offers valuable insights.

7.1 Fast Fourier Transform for pricing

In this section, we briefly describe the application of the Fast Fourier method, as proposed in Carr and Madan Carr [6] to our models. Building upon the discussion in Section 3, the exponential model for forward prices is:

$$\log F(t,T_1,T_2) = \log F(0,T_1,T_2) + h(t,T_1,T_2) + X + Y$$

where $X = \Gamma_2(T_1,T_2)W_2(t) + \Sigma_D(t,T_1,T_2)Z_D(t)$ and $Y = \Sigma_J(t,T_1,T_2)Z_J(t)$. Moreover, let $E = X+Y$. To utilize the Fast Fourier Transform (FFT), we must calculate the characteristic function of log-prices, denoted as $\Phi(u) = \mathbb{E}\left[e^{iu\log F(t,T_1,T_2)}\right]$, which incidentally computes the characteristic function of E.

$$\phi_E(u) = \mathbb{E}\left[e^{iuE}\right] = \mathbb{E}\left[e^{iuX}\right]\mathbb{E}\left[e^{iu\Sigma_D(t,T_1,T_2)}\right] = \phi_X(u)\phi_{Z_{J(t)}}(u\Sigma_D(t,T_1,T_2)).$$

Since $X \sim \mathcal{N}\left(0, \sigma^2(t,T_1,T_2)\right)$, with $\sigma^2(t,T_1,T_2) = \Sigma_D^2(t,T_1,T_2)\left(\frac{1-e^{-2k_D t}}{2k_D}\right) + \Gamma_2(T_1,T_2)^2$ we have:

$$\phi_E(u) = e^{-\frac{1}{2}\sigma^2(t,T_1,T_2)u}\phi_{Z_{J(t)}}\left(u\Sigma_D(t,T_1,T_2)\right).$$

Therefore, all we need to do is compute the characteristic function of:

$$Z_J(t) = \sum_{n=1}^{N(t)} J_n e^{-k_J(t-\tau_n)}.$$

and evaluate it at $u\Sigma_D(t,T_1,T_2)$. From Theorem 3.3 in Sabino and Cufaro Petroni [34] it is easy to show that:

$$\phi_{Z_{J(t)}}(u) = \left(\frac{\lambda_u - iue^{-k_J t}}{\lambda_u - iu}\right)^{p\lambda/k}\left(\frac{\lambda_d + iue^{-k_J t}}{\lambda_d + iu}\right)^{(1-p)\lambda/k},$$

and hence $\phi_E(u)$ is obtained.

To ensure that forward prices are martingales, we have to impose a condition on $h(t,T_1,T_2)$, which is given by:

$$h(t,T_1,T_2) = -\log\phi_E(-i).$$

Once $\phi_E(u)$ and $h(t,T_1,T_2)$ are determined,, the price of the call option $C(t,T,T_1,T_2,k)$ with maturity T and log-strike k can be computed as:

$$C(t,T,T_1,T_2,k) = \frac{e^{\alpha k}}{\pi}\int_0^\infty \frac{e^{-rT}\Phi(v-(\alpha+1)i)}{\alpha^2+\alpha-v^2+i(2\alpha+1)v}, \tag{60}$$

where α is the dumping factor introduced in Carr and Madan Carr [6] and $\Phi(u)$ is the Fourier transform of the log-price $\log F(T,T_1,T_2)$ at time T, and it is given by:

$$\Phi(u) = e^{iu(\log F(0,T_1,T_2)+h(T,T_1,T_2))}\phi_E(u).$$

The integral in Equation (60) can be efficiently approximated using the FFT algorithm by Cooley and Tukey [8]. Therefore, to employ the models we presented, one simply needs to compute the characteristic function $\Phi(u)$ of log-prices.

Alternatively, if we aim to price using Fourier methods with multiplicative NOA models for forward prices of the form

$$\begin{aligned} F(t,T_1,T_2) =& F(0,T_1,T_2)(1+h(t,T_1,T_2)+\Gamma_2(T_1,T_2)W_2(t) \\ &+\Sigma_D(t,T_1,T_2)Z_D(t)+\Sigma_J(t,T_1,T_2)Z_J(t)) \end{aligned}$$

The approach we essentially employ is based on the valuation of the so-called *time-value*, as proposed in Piccirilli et al. [26]. Similar to the exponential case, we need to ensure that forward prices are martingales, thus requiring:

$$h(t,T_1,T_2) = -\Sigma_J(t,T_1,T_2)\mathbb{E}\left[Z_J(t)\right].$$

The expected value of $Z_J(t)$ can be readily computed by beginning with the characteristic function and recalling the relationship between the characteristic function $\phi_X(u)$ of the random variable X and its moment generating function $m_X(u)$:

$$m_X(u) = \phi_X(-iu).$$

To assess the reliability of the proposed algorithms across diverse market conditions, we compare the results in Table 8 and 9, obtained using the FFT method and Monte Carlo simulation with identical parameter sets as detailed in Section 6. We observe closely matching results from both methods, indicating their suitability for pricing tasks. Particularly noteworthy, as widely documented in literature, is the computational efficiency of Fourier Transformation-based methods compared to Monte Carlo techniques, making them preferable for calibration routines wherein pricing formulas are invoked numerous times during numerical optimization processes.

Table 8: Double exponential jumps, exponential setting.

	MC	FFT	MC	FFT	MC	FFT	MC	FFT
$K\backslash T_0$	0.25		0.5		0.75		1	
85	18.47	18.41	22.13	22.07	25.88	25.78	29.88	29.95
90	14.76	14.70	18.95	18.91	23.04	22.95	27.30	27.36
95	11.44	11.38	16.07	16.03	20.42	20.34	24.90	24.96
100	8.58	8.52	13.48	13.45	18.03	17.95	22.67	22.72
105	6.23	6.17	11.20	11.18	15.86	15.78	20.62	20.66
110	4.38	4.33	9.22	9.19	13.90	13.82	18.73	18.76
115	2.99	2.96	7.52	7.49	12.14	12.07	16.99	17.02
120	1.99	1.97	6.08	6.06	10.57	10.50	15.40	15.42

Carr and Madan [6] pioneered option pricing via characteristic functions, though their approach required a damping factor. Lewis [23] extended this by using complex

Table 9: Double exponential jumps, NOA setting.

	MC	FFT	MC	FFT	MC	FFT	MC	FFT
$K\backslash T_0$	0.25		0.5		0.75		1	
85	19.64	19.60	24.25	24.24	29.04	29.05	34.29	34.53
90	15.81	15.78	20.80	20.81	25.79	25.81	31.16	31.40
95	12.30	12.26	17.59	17.60	22.72	22.75	28.18	28.41
100	9.18	9.15	14.63	14.65	19.84	19.87	25.35	25.58
105	6.54	6.51	11.95	11.98	17.16	17.20	22.68	22.90
110	4.41	4.40	9.57	9.61	14.70	14.74	20.16	20.38
115	2.81	2.81	7.50	7.54	12.45	12.49	17.82	18.02
120	1.69	1.70	5.74	5.78	10.42	10.47	15.64	15.84

plane integration to avoid this issue, whereas Fang and Oosterlee [11] developed the COS method, using a cosine series from characteristic functions, yielding high accuracy and speed. The recent PROJ algorithm (see Kirkby [18]) further improved performance by using Riesz bases, showing robustness and efficiency compared to COS. Overall, characteristic functions enable many pricing methods; in this work, we use Carr and Madan's method, given its reliability and acceptance for benchmarking against Monte Carlo simulations.

7.2 Single underlying contract calibration versus whole market

By using dynamics in Equations (17) and (19) we can effectively model either individual underlying contracts or, alternatively, assume that the entire market is influenced by two stochastic factors, as proposed, for instance, in Kiesel et al. [17] and Piccirilli et al. [26]. In the former scenario, we assume that at a given time t, N prices $C(t, T_n, T_1, T_2, K_n)$ of call options with maturities T_n and strike prices K_n, written on forwards with deliveries within $[T_1, T_2]$, are quoted in the market. As is customary in practice, see Cont and Tankov [7], to fit the model on observed prices, we seek a set of parameters that minimizes a designated error function. Specifically, our objective is to solve:

$$\Theta^* = \arg\min_{\boldsymbol{\Theta}} \sum_{n=1}^{N} \omega_n \left(C(t, T_n, T_1, T_2, K_n) - C_{\boldsymbol{\Theta}}(t, T_n, T_1, T_2, K_n) \right)^2, \tag{61}$$

where $\boldsymbol{\Theta}$ is the vector parameters, $C_{\boldsymbol{\Theta}}(t, T_n, T_1, T_2, K_n)$ are option prices obtained from the model and ω_n denotes the weights that reflect our confidence in individual data points, which is often related to the liquidity of the corresponding derivatives. As suggested by Cont and Tankov [7], a common choice is:

$$\omega_n = \frac{1}{|C_n^{bid} - C_n^{ask}|},$$

which assigns higher weight to more liquid instruments with narrower bid-ask spreads. Alternative weighting schemes are also possible, for example, using a function of moneyness to emphasize at-the-money options, which typically carry more information about implied volatility. For simplicity, however, we assume equally weighted data by setting $\omega_n = 1$.

The option price $C_{\mathbf{\Theta}}(t, T_n, T_1, T_2, K_n)$ in Equation (61) can be computed either using a closed-form expression or via a Fourier-based method. Both approaches have their advantages and disadvantages. Although closed-form solutions are generally more efficient and numerically stable than numerical methods, a Fourier-based approach may be preferable when pricing options across many strikes for a single maturity. This is because the FFT algorithm allows for the rapid computation of a full panel of derivative prices. Moreover, FFT-based methods are commonly used in the industry because they are often more flexible and easier to maintain than implementing multiple analytical formulas, each tailored to a specific pricing model.

If we consider the presence of M forward contracts with delivery periods $[T_{m,1}, T_{m,2}]$, where $m = 1, \ldots, M$, one can conjecture that the entire market is influenced by only two principal risk factors. Thus, assuming that:

$$\begin{aligned} \log F(t, T_{m,1}, T_{m,2}) = &\log F(0, T_{m,1}, T_{m,2}) + h_m(t, T_{m,1}, T_{m,1}) + \Gamma_2(T_{m,1}, T_{m,2}) W_2(t) \\ &+ \Sigma_D(t, T_{m,1}, T_{m,2}) Z_D(t) \\ &+ \Sigma_J(t, T_{m,1}, T_{m,2}) Z_J(t), \quad m = 1, \ldots, M. \end{aligned} \tag{62}$$

A similar dynamics holds for the NOA model according to Equations (19).

In power markets, options are typically traded on forward contracts with varying delivery lengths, some of which may overlap. For instance, contracts for Jan/YY, Feb/YY, Mar/YY, and Q1/YY may be simultaneously traded. As shown in Latini et al. [21] and Piccirilli et al. [26], to ensure the consistent estimation of parameters and prevent arbitrage opportunities, additional constraints on volatility parameters are necessary. For instance, in the case of the NOA model, constraints such as those derived in Piccirilli et al. [26] can be applied to the volatility components and to $h_m(t, T_{m,1}, T_{m,2})$. Conversely, when utilizing the approach outlined in Equation 62, imposing constraints on parameters becomes challenging. Such constraints are typically essential only for overlapping products; thus, from a practical standpoint, fully overlapping products are excluded from the calibration process. Consequently, the least-squares problem is:

$$\mathbf{\Theta}^* = \arg\min_{\Theta} \sum_{m=1}^{M} \sum_{n=1}^{N} \left(C(t, T_{m,n}, T_{m,1}, T_{m,2}, K_{m,n}) - C_\Theta(t, T_{m,n}, T_{m,1}, T_{m,2}, K_{m,n}) \right)^2,$$

with $T_{m,n} < T_{m,1}$. In the following section, we implement the calibration procedure outlined above using real market data. Subsequently, we compare the obtained results with those of a model benchmark.

7.3 Real world application

In this section we apply the aforementioned models to real market date, with a specific focus on the German power futures market. To ensure clarity in the dissertation, we concentrate solely on the case of a single underlying contract. Extending the analysis to encompass multiple underlying contracts would be straightforward and analogous to the methodology outlined in Piccirilli et al. [26].

In our application, we focus on European call options traded on the EEX Power Derivatives market. The underlying asset is a futures contract delivering 1 MWh for each hour within the delivery period $[T_1, T_2]$. Specifically, we examine options written on the Base index of the German area with a delivery period in the year 2025, known as Phelix Base yearly options. Typically, these options offer various maturities, including March 24, June 26, September 25, and December 12, 2024. For our experiment, we only consider settlement prices, as they provide valuable insights into market expectations, despite not always representing actual trades. To focus on options closer to the money, we restrict our analysis to strike prices within ± 25

For the trading date, we select February 5, 2024, where the settlement price $F(t, T_1, T_2)$ of the forward contract delivering in 2025 is EUR 79.00 per MWh. Lastly, we use a risk-free rate of $r = 0.04$, aligning with the prevailing economic conditions in Europe.

As stated at the beginning of this section, we focus on the Kou model introduced in Section 3 both in exponential and NOA form in which, for simplicity, we assume that the function $\gamma_2(t)$ is constant in time and equal to $\gamma_2 > 0$. In this case the vector of unknown parameters is given by $\mathbf{\Theta} = (\gamma_2, \sigma_d, k_d, \gamma_j, k_j, \lambda_p, p, \lambda_d, \lambda_u, \gamma_2)$. All the other models can be considered and the calibration procedure remains the same.

To facilitate a comparison between the Kou's model and a conventional financial model, we employ a Variance Gamma-like model obtained by combining the variance gamma process introduced by Madan, Carr and Chang [24] and the OU-VG procces studied by Sabino [29]. The futures prices can be modeled as follows:

$$F(t, T_1, T_2) = F(0, T_{1,i}, T_{2,i}) e^{h(t,T_1,T_2) + Y_1(t) + \int_0^t e^{-k(t-u)} dY_2(u)} \tag{63}$$

where $Y_1 = \{Y_1(t); t \geq 0\}$ is a Variance Gamma process with parameters (μ_1, σ_1, ν_1) and $Y_2 = \{Y_2(t); t \geq 0\}$ is another Variance Gamma process, independent from Y_1, with parameters vector (μ_2, σ_2, ν_2) and $k > 0$. In this case the vector of parameters is given by $\mathbf{\Theta} = (\mu_1, \sigma_1, \nu_1, \mu_2, \sigma_2, \nu_2, k)$.

Under the same assumptions for Y_1 and Y_2 a NOA approach can be adopted. It is possible to model the forward price $F(t, T_1, T_2)$ as:

$$F(t, T_1, T_2) = F(0, T_{1,i}, T_{2,i}) \left(1 + h(t, T_1, T_2) + Y_1(t) + \int_0^t e^{-k(t-u)} dY_2(u) \right).$$

In both scenarios, the function $h(t, T_1, T_2)$ must be selected to ensure that futures prices behave as martingales, employing the rationale discussed in Section 7.1. The characteristic function of the OU-VG process at time t, denoted by the term $\int_0^t e^{-t(t-u)} dY_2(u)$, can be computed using the approach outlined in [Proposition 3.1] Sabino [29].

Under this framework, we can address the minimization problem in Equation (61) for all four models under consideration.

Solving the minimization problem outlined in Equation (61) requires numerous evaluations of the objective function. Consequently, efficient methods for assessing the option price $C_{\mathbf{\Theta}}(t, T_n, T_1, T_2, K_n)$ are crucial. It is customary to utilize closed-form solutions whenever available. Alternatively, methods based on the Fourier transform, briefly introduced at the outset of Section 7.1, can be employed.

In Tables 10, 11, 12 and 13 we present the calibrated parameters for the chosen data-set corresponding to the four proposed models. Figure 1 illustrates a comparison between the option prices derived from the models and those quoted in the market. Our observations indicate that all models successfully replicate the observed market option prices. Examining the root-mean-square-errors (RMSE) defined as:

$$RMSE = \sqrt{\frac{\sum_{n=1}^{N} \left(C_{\mathbf{\Theta}^*}(t, T_n, T_1, T_2, K_n) - C(t, T_n, T_1, T_2, K_n)\right)^2}{N}},$$

in Table 14, we find that the Exponential Kou model provides the best fit, while the NOA versions perform least favorably. Variance gamma exponential model exhibits intermediate performance. The superior fit achieved by the Kou exponential model is attributed to its higher number of parameters, which increases the complexity of the calibration task. Generally, it is advisable to favor models with fewer parameters whenever possible. In this context, all the model emerge as a reasonable choices for modeling futures prices $F(t, T_1, T_2)$ since the implied volatility surface is relatively uncomplicated. However, for more exotic shapes in the implied volatility structure, models with more parameters, such as those presented in the previous sections, might yield improved results. This scenario could occur when dealing with more liquid markets than the futures power one, such as the natural gas market, where a greater number of options are traded. This prospect constitutes an interesting avenue for future research and will be explored in subsequent inquiries.

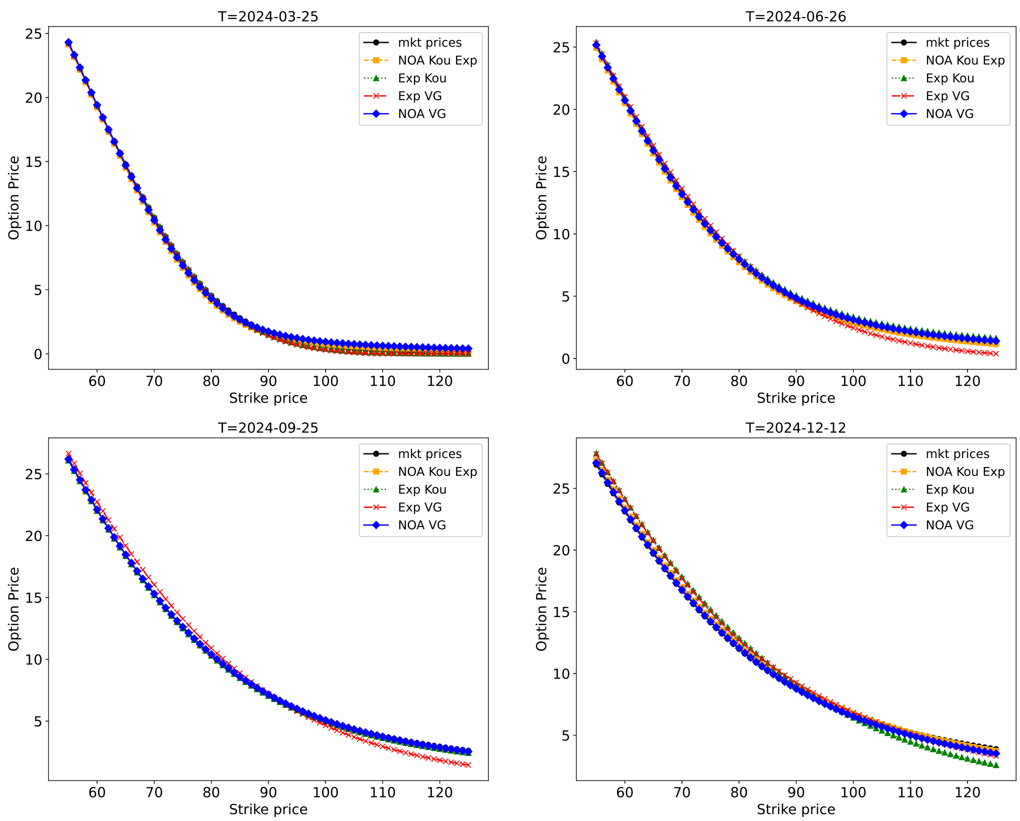

Figure 1: Market repricing with different models of call options written on Calendar 2025 with four different maturities.

Parameter	Value	Parameter	Value
γ_2	0.4159	γ_D	0.0828
k_D	0.3969	γ_J	0.0839
k_J	0.6873	λ_P	0.6900
λ_d	2.1811	p	0.5534
λ_u	5.3616		

Table 10: Parameters estimated for the exponential Kou model with double exponential jumps.

Parameter	Value	Parameter	Value
γ_2	0.3068	γ_D	0.2335
k_D	0.4354	γ_J	0.9750
k_J	0.7813	λ_P	0.5852
λ_d	1.8456	p	0.5811
λ_u	6.3699		

Table 11: Parameters estimated for the NOA - Kou model with double exponential jumps.

Parameter	Value	Parameter	Value
μ_1	0.1716	μ_2	-0.7313
σ_1	0.1356	σ_2	0.3370
ν_1	1.5870	ν_2	0.0124
k	0.4740		

Table 12: Parameters estimated for the exponential Variance Gamma model.

Parameter	Value	Parameter	Value
μ_1	0.2457	μ_2	-0.7514
σ_1	0.3189	σ_2	0.2791
ν_1	1.5970	ν_2	0.0532
k	0.4962		

Table 13: Parameters estimated for the multiplicative Variance Gamma model.

Model	RMSE
Kou Exponential	0.4664
Kou NOA	0.1867
Exponential variance gamma	0.1579
NOA variance gamma	0.5491

Table 14: RMSE for the different models.

8 Conclusions and future inquiries

In conclusion, our study has delved into various facets of option pricing and modeling within energy markets. We introduced enhancements to the established Kou model, addressing the challenge of accommodating the Samuelson effect and other non-Gaussian features prevalent in real market data. By expanding upon this model, we derived closed-form solutions for vanilla options, simplifying previously complex formulas and extending their usability.

Our contributions encompass multiple aspects, from determining convolution densities to providing closed-form solutions for vanilla options. The derived formulas avoid numerical integration or approximation, ensuring accuracy and computational efficiency.

Furthermore, our exploration and empirical validation of the extended Kou model against observed implied volatilities and the variance gamma model demonstrated its robustness and applicability to real-world scenarios.

While our study presents promising results, certain areas warrant further investigation. The complexity involved in parameter estimation calls for more nuanced techniques beyond standard log-likelihood methods, which we highlighted in our findings.

While not explicitly articulated in this paper for the sake of brevity, many of the outcomes obtained can be directly applied to market models reliant on Γ-modified-Gaussian-Ornstein-Uhlenbeck (OU) processes, including their bilateral counterparts, frequently employed within energy markets. It is easy to verify that these models exhibit a transition density which is a mixture of Erlang-like laws with a Pólya mixing distribution. Consequently, parameter estimation, often solely reliant on historical data, requires suitable adaptations such as employing the EM method or stochastic descent. This approach stands to offer superior solutions compared to existing methods based on Gibbs sampling and Markov Chain Monte Carlo.

In summary, our research contributes to advancing the understanding and practical implementation of option pricing models tailored to energy markets. Through empirical validations, novel methodologies, and the derivation of closed-form solutions, our research aims to provide a deeper understanding of non-Gaussian processes in energy markets.

Disclosure of interest

The authors declare that they have no conflicts of interest relevant to the content of this article.

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